SOIL EROSION AND CONSERVATION · NOTE ER10

Wind Erosion — Mechanics, Soil Movement, Erodibility Factors and the Wind Erosion Equation

Course topics: CAE1.3 L9, SWCE-501 T7, SWCE-505 T3

What wind erosion is

Wind erosion is the detachment, transport and deposition of soil by wind. Soil may be removed and redeposited in uniform layers or as local blowouts and sand dunes.

- It removes silt, clay and organic matter, lowering productivity and water-holding capacity and leaving coarse, sandy, less organic soil that erodes even more easily.
- It damages crops — especially seedlings, by abrasion, root exposure and burial — spreads weeds and insects, buries shrubs, fences, wells, ditches and channels, and blocks roads and railways, raising maintenance costs.

Wind erosion in India

- About **17.2 million ha** is affected by wind erosion and desert extension — worst in western Rajasthan and adjoining areas. The Rajasthan desert covers **2,08,000 km²** and extends into Haryana, Punjab and Gujarat. Climate: high temperature, strong wind, low rainfall, low relative humidity.
- **Western Rajasthan:** annual rainfall 75–450 mm, ~90% of it in 2–6 monsoon showers. Mean wind speed 3.2–32 km/h, generally from the south-west, with high winds in May–June when the soil is driest.
- **Land-use causes:** extensive cultivation and heavy grazing; frequent up-and-down tillage (no contour farming); removal of crop residues (loss of mulch); lopping trees for fodder in summer, exposing the soil.

How wind moves soil

- Wind acts parallel to or against the surface. Its speed is lowest near the ground and increases with the logarithm of height. Grains move when wind pressure overcomes gravity and inertia.
- Three stages: **initiation** of movement → **transport** (through the air or along the surface) → **deposition** elsewhere.

Mode	Particle size	Mechanism	Share of soil moved
Saltation	0.05–0.5 mm	The main mode — a series of bounces. Wind pressure and collisions lift grains: a short roll, a jump into the air, then a fall and rebound. The jump rises about 1/5 to 1/4 of its horizontal length (L) and descends at 6°–12°	50–75%
Suspension	< 0.05 mm, especially < 0.01 mm	Fine particles float in the air stream, lifted mainly by the impact of saltating grains (very fine dust resists direct lift by the wind because of cohesion and its submergence in the turbulent air); upward eddies can carry it many kilometres	3–40%
Surface creep	0.5–2 mm	Large particles roll or slide, pushed mainly by saltating grains rather than directly by wind	5–25%

- Coarse soils move mostly by surface creep; pulverised soils by saltation and suspension.
- **~93%** of wind-borne soil travels below **30 cm** height.

Sand dunes

- A sand dune is a wind-built sand wave of roughly triangular section. Once exposed, sand moves with the prevailing wind; the threshold velocity is lower on dunes than on uneroded fields.
- **Slopes:** windward 5–10° (the notes add 5–11%); leeward ~30° (the notes add 33%).
- Crescent-shaped dunes advance as grains move from the windward to the leeward face, shifting several metres in a severe windstorm. Large dunes destroy land, crops, buildings and forests.

Check: the bracketed percentages do not match the angles. As gradients, 5–10° ≈ 9–18% and 30° ≈ 58% (tan θ). Use the angles.

Threshold velocity

- The minimum wind speed that starts particles moving. **Fluid threshold:** movement by direct wind action. **Impact threshold:** movement by the impact of saltating grains.
- For ~0.1 mm particles it is about **16 km/h at 30 cm** height. It rises with size above 0.15 mm, and also rises as size falls below 0.05 mm.

Factors controlling wind erosion

Group	Factors
Wind	Speed, direction, structure (turbulence), temperature, humidity and the load it carries
Surface	Roughness, plant cover and obstructions
Topography	Flat, undulating or broken
Soil	Texture, structure, organic matter and moisture

Of the controllable factors, **soil cloddiness, surface roughness and crop residues** account for about 75% of the variation in erosion.

- **Texture and structure:** particles < 1 mm dominate most farm soils. Coarse sandy loam, loamy sand and sand are most susceptible — little silt and clay to bind them, so single-grain structure or weak clods. The coarsest and finest soils are the most erodible; loam, silt loam, clay loam and silty clay loam resist best.

- **Crusts and clods:** consolidation into crusts or clods forms large stable units that resist erosion. Rain disperses the soil and on drying it forms a thin compact crust, usually ≤ 1.5 mm thick. Crust and clod stability depends on moisture, compaction, organic matter, clay, lime, micro-organisms, pH and cementing materials. A cloddy surface slows the wind and traps soil.
- **Particle size and density:** erodibility depends on the proportion, size and bulk density of erodible particles. Equivalent diameter = ρD/2.65 (ρ = particle density, D = diameter). The most erodible particles are about **0.1 mm** equivalent diameter; larger and smaller particles resist more. Erodible fractions keep blowing away until the non-erodible fraction shelters the surface.
- **Surface roughness:** rough surfaces slow the wind and trap moving grains. Roughness depends on the height and density of vegetation; the size, shape and spacing of clods, ripples and ridges; and the tillage method and implement. Alternate ridges and furrows trap saltating grains and reduce erosion downwind — provided the ridge tops are cloddy enough.
- **Vegetation and residues:** living or dead plants slow the wind at the surface, absorb its force and trap drifting soil. Standing, firmly anchored vegetation works best, and vegetative protection is the most reliable permanent control. Excessive tillage, wrong implements and overgrazing strip it away.

Correction: the notes say "particles larger and smaller than 1 mm are more resistant", which contradicts the 0.1 mm optimum given just before. It should read 0.1 mm (Chepil).

Estimating wind erosion

Wind erosion equation: $E = f(I, K, C, L, V)$

Symbol	Meaning
E	Estimated average annual soil loss (t/ha/year)
I	Soil erodibility index (Mg/ha/year) — a function of the soil aggregates larger than 0.84 mm
K	Ridge roughness factor
C	Climate factor — climatic erosivity from wind speed and surface moisture
L	Unsheltered field length (m) — along the prevailing wind from the sheltered edge to the end of the field
V	Vegetative cover factor — type, amount and orientation of vegetation

The factors are inter-dependent and must be combined, not simply multiplied, to estimate erosion.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Woodruff, N. P. and Siddoway, F. H. (1965), A wind erosion equation, Soil Sci. Soc. Am. Proc. 29: 602–608. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The main mode of wind-borne soil movement is

(a) suspension
(b) saltation
(c) surface creep
(d) rolling

Q2. Saltating particles are about

(a) < 0.01 mm
(b) 0.05–0.5 mm
(c) 0.5–2 mm
(d) > 2 mm

Q3. About 93% of wind-blown soil moves below a height of

(a) 5 cm (b) 30 cm
(c) 1 m (d) 3 m

Q4. Threshold wind velocity for 0.1 mm particles is about

(a) 5 km/h (b) 16 km/h at 30 cm
(c) 40 km/h (d) 80 km/h

Q5. Most erodible particles by wind have an equivalent diameter of about

(a) 0.01 mm (b) 0.1 mm
(c) 1 mm (d) 2 mm

Q6. In the wind erosion equation, I depends on aggregates larger than

(a) 0.1 mm (b) 0.42 mm
(c) 0.84 mm (d) 2 mm

Q7. Surface creep particles are

(a) < 0.05 mm (b) 0.05–0.5 mm
(c) 0.5–2 mm (d) 2–5 mm

Q8. Annual rainfall in western Rajasthan is

(a) 75–450 mm (b) 450–750 mm
(c) 750–1000 mm (d) > 1000 mm

Q9. Which soil resists wind erosion best?

(a) sand (b) loamy sand
(c) silt loam (d) coarse sandy loam

Answers

Q1 — (b) saltation — *50–75% of soil moved.*
Q2 — (b) 0.05–0.5 mm
Q3 — (b) 30 cm
Q4 — (b) 16 km/h at 30 cm
Q5 — (b) 0.1 mm
Q6 — (c) 0.84 mm
Q7 — (c) 0.5–2 mm
Q8 — (a) 75–450 mm
Q9 — (c) silt loam